Jenkins Freestyle Project - Practical Tasks

Task 1: Create a Simple Hello World Project

• Create a new Freestyle project called HelloWorldJob.
• Add a Windows batch command or Shell command (Linux) to print 'Hello, Jenkins!'.
• Run the job and check the Console Output.
Solution:
```
Windows:
echo Hello, Jenkins!

Linux:
echo "Hello, Jenkins!"
```

Task 2: Use Workspace Variable

• Create a project called WorkspaceJob.
• Add a build step that prints the WORKSPACE environment variable.
• Verify in the console output that the path to the workspace is correct.
Solution:
```
Windows:
echo The workspace path is: %WORKSPACE%

Linux:
echo "The workspace path is: $WORKSPACE"
```

Task 3: File Creation and Copy

• Create a project FileCopyJob.
• Add a batch command that creates a file test.txt in the workspace and copies it to a folder.
• Verify the file is copied.
Solution:
```
Windows:
echo Hello from Jenkins > %WORKSPACE%\test.txt
copy %WORKSPACE%\test.txt C:\inetpub\wwwroot\Devops\

Linux:
echo "Hello from Jenkins" > $WORKSPACE/test.txt
cp $WORKSPACE/test.txt /var/www/devops/
```

Task 4: Parameterized Build

• Create a project ParameterizedJob.
• Add a String Parameter called USERNAME.
• In the build step, print Hello, $USERNAME.
• Build the job multiple times with different values and check the output.
Solution:
```
Windows:
```

```
echo Hello, %USERNAME%

Linux:
echo "Hello, $USERNAME"
```

Task 5: Schedule Builds

• Create a project ScheduledJob.
• Configure it to run every 2 minutes (H/2 * * * *).
• In the build step, echo the current date and time.
• Verify multiple builds are triggered automatically.
Solution:

```
Windows:
echo Current date and time: %date% %time%

Linux:
echo "Current date and time: $(date)"
```

Task 6: Trigger by SCM (Git)

• Create a project GitJob.
• Connect it to a GitHub repository.
• Configure to pull the latest code.
• In the build step, print the last commit message.
• Verify Jenkins fetches and displays commits.
Solution:

```
Linux/Windows (Git Bash):
git log -1 --pretty=%B
```